Irrigation Source Development Optimization

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Introduction: The goal of this analysis is to find the optimal way to develop sources of water for irrigation to maximize profit. A river supplies water for three potential projects. A dam (either high or low) can be built to create a reservoir with a gravity diversion. There is also opportunity for a pump downstream of the potential reservoir location. This analysis considers the capital, operational costs, and potential earnings from each project over the course of a year (two seasons). In this report, I will describe the model I used and the optimal solution I found for which projects should be built and how much water they should supply each season. I will also give my recommendations and insights from the solution.

Model Formulation: The information present in the problem can be sorted based on two dimensions (or categories): where the water is sourced from (the high dam reservoir, low dam reservoir, or the water pump) and when the water is sourced (season 1 or season 2). The decisions that will affect the profit are whether to build each project and how much water to source from each project each season. The profit is the sum of the revenue generated minus the costs. The revenue in this case is the total volume of water sourced multiplied by the rate of acres able to be irrigated per ac-ft of water and the revenue generated per irrigated acre. There are varying capital costs associated with building each project. There is also an operational cost per ac-ft of water p

Solution:

Insights and Recommendations:

Conclusion:

AI Usage:

References:

Appendix:

Dimensions: Water sources (source) = {high dam, low dam, river diversion}, Time (t) = {1,2}. And Water Stores (storage) = {reservoir storage}

Decision Variables: What water sources to build and how much water to source from each location in each season

$$I_{source}\ [0\ =\ not\ built, 1\ =\ build]$$

$$X_{source,t}\ [acft]$$

Objective: The total profit is the revenue generated per acre irrigated multiplied by the number of the acres irrigated minus the costs to develop the water sources and to operate them.

$$Max\ P\ = \sum_{source,t} Rev * d_t * X_{source,t} - \sum_{source} cc_{source} * I_{source} - \sum_{source,t} c_{source} * X_{source,t}$$

d_t = irrigation demands per acre at each time step [acft/acre]

cc_{source} = capital cost to develop each source [$]

c_{source} = cost per acft of water from each source [$/acft]

Constraints:

The water extracted from each source must be less than the capacity if the capital cost has been paid. If the capital cost has not been paid, no water can be extracted from the source

$$X_{source,t} \leq\ cap_{source} *\ I_{source}$$

Only one dam can be built

$$I_{high\ dam}\ +\ I_{low\ dam} \leq\ 1$$

Mass balance – The maximum water that can be extracted from the river diversion is the Inflow minus the water diverted from the reservoir plus the groundwater recharge

$$X_{river,t} \leq\ Q_t\ -\ X_{high\ dam,t}\ -\ X_{low\ dam,t}\ +\ G$$